Comparing the motions of a protein and its mutants

Notes for penm

2026-10-16

1 Scope

These notes collect a review of penm's `delta_motion_*` measures, the measures used in the literature on comparing protein motions, and a study of how a protein's normal modes change under single-point mutations. Everything numerical was computed with the scripts in this folder; each script's printed output is saved beside it as `<script>.out.txt`. The protein is 2acy throughout, and the results come from one protein, one model and one ensemble: whether they generalise is not known.

2 The motion measures in penm

Let $\lambda_n, \mathbf{u}_n$ be the eigenvalues and eigenvectors of the wild type's network matrix $\mathbf{K}$ over its $3N-6$ internal modes, $\mu_m, \mathbf{w}_m$ those of the mutant, $\mathbf{C} = \beta^{-1}\mathbf{K}^{+}$ the covariance, $\sigma_n^2 = 1/(\beta\lambda_n)$, and $O_{nm} = \mathbf{u}_n \cdot \mathbf{w}_m$ the overlap matrix. $\mathbf{C}_{ii}$ is the 3×3 diagonal block of site i.

function	quantity	depends on
`dmsfi`	$\operatorname{tr}\mathbf{C}_{ii}^{\mathrm{mut}} - \operatorname{tr}\mathbf{C}_{ii}^{\mathrm{wt}}$	site marginals
`dbhati`	Bhattacharyya distance between the zero-mean Gaussians with $\mathbf{C}_{ii}^{\mathrm{wt}}$, $\mathbf{C}_{ii}^{\mathrm{mut}}$	site marginals
`rwsipi`	RWSIP of $\mathbf{C}_{ii}^{\mathrm{wt}}$ and $\mathbf{C}_{ii}^{\mathrm{mut}}$	site marginals
`dhi`	$\frac{1}{2}\ln\det\mathbf{C}_{ii}^{\mathrm{mut}} / \det\mathbf{C}_{ii}^{\mathrm{wt}}$	site marginals
`dmsfn`	$\mathbf{u}_n^T\mathbf{C}^{\mathrm{mut}}\mathbf{u}_n - \sigma_n^2$	wt modes
`dhn`	$\frac{1}{2}\ln\left(\mathbf{u}_n^T\mathbf{C}^{\mathrm{mut}}\mathbf{u}_n/\sigma_n^2\right)$	wt modes
`rwsipn`	$\left(\sigma_n^2\sum_m \mu_m^{-1}\beta^{-1}O_{nm}^2 \big/ \sum_k \sigma_k^2\beta^{-1}\mu_k^{-1}\right)^{1/2}$	both mode sets
`nhn`	$\exp\left(-\sum_m O_{nm}^2 \ln O_{nm}^2\right)$	both mode sets

The four site measures use only the 3×3 diagonal blocks, the marginal distribution of each site: none of them sees a change in the correlations between sites.

2.1 Identities

Checked on one genm mutant (site 11, letter 1; `motion_identities.R`):

- $\texttt{dhn}_n = \frac{1}{2}\ln(1 + \texttt{dmsfn}_n/\sigma_n^2)$ exactly.
- $\texttt{rwsipn}_n = \left(\sigma_n^2(\sigma_n^2 + \texttt{dmsfn}_n)/\sum_k \sigma_k^2\mu'_k\right)^{1/2}$ exactly, with μ'_k the mutant's variances paired with the wild type's by rank, because $\sum_m O_{nm}^2/(\beta\mu_m) = \mathbf{u}_n^T\mathbf{C}^{\mathrm{mut}}\mathbf{u}_n$. Given the wild type, `dhn` and `rwsipn` carry the same information as `dmsfn`.

- $\sum_n \texttt{dhn}_n = -0.099$, while the entropy change is $\Delta S/k = -0.189 = -\beta\,\texttt{ddg_tds}$. The two differ by $\frac{1}{2}\left(\sum_n \ln[\mathbf{C}']_{nn} - \ln\det\mathbf{C}'\right) \geq 0$, the multi-information of the mutant's covariance $\mathbf{C}'$ in the wild type's mode basis, so the `dhn` do not decompose the entropy change.
- `rwsipi` ranges over 0.9996–1 for this mutant.

3 Measures in the literature

The papers are in `dev/docs/references/comparison`. All compare homologous proteins, which needs alignment, a common structural core, superposition and, usually, normalisation of the amplitudes; penm compares a protein with mutants built in the same network, on the same sites.

- **Maguid et al. 2005** (Biophys. J. 89:3). GNM modes of 18 globins, restricted to gap-free positions; modes matched across proteins by permuting the columns of the overlap matrix to maximise its trace; representative modes by SVD; Pearson r of B-factor profiles.
- **Maguid et al. 2006** (J. Mol. Evol. 63:448). Spearman ρ_B of experimental C_α B-factor profiles of homologous pairs, against random pairs, with RMSD and sequence identity.
- **Maguid et al. 2008** (Gene 422:7). GNM. Mode conservation S_{nn}, the diagonal of the overlap matrix after reassignment; collectivity $\kappa_n = N^{-1}\exp(-\sum_i q_{in}^2 \ln q_{in}^2)$; Spearman correlation of B-factor profiles. Conservation decreases with mode index and correlates with collectivity.
- **Fuglebakk, Echave & Reuter 2012** (Bioinformatics 28:2431). A score is a property ($\mathbf{C}$, the correlation matrix $\mathbf{P}$, the isotropic $\mathbf{C}_{\text{iso}}$ and $\mathbf{P}_{\text{iso}}$, the RMSF profile) and a measure (RMSIP over 10 modes, RWSIP, and a normalised Bhattacharyya distance NDB between zero-mean Gaussians after trace normalisation and projection onto a common subspace). Judged by agreement with SCOP: the property matters more than the measure; RMSF profiles do worst; NDB($\mathbf{C}$) and RWSIP($\mathbf{P}$) do best, rivalling RMSD.
- **Micheletti 2013** (Phys. Life Rev. 10:1, review). RMSIP; a per-residue profile Q_i of it; the distance-weighted RMSIP of Zen et al. 2008, which combines structure and motion by weighting each aligned residue with $f(d_i) = [1 - \tanh((d_i - 4\,\text{Å})/2\,\text{Å})]/2$; matrices of distance fluctuations (Münz et al.).
- **Echave 2013** (comment on the review). The low modes may be conserved because they are robust to random mutations, not because selection keeps them.
- **Granata et al. 2017** (PNAS 114:E10612). Compares partitions into domains, coevolutionary against SPECTRUS dynamical domains, by adjusted mutual information.

The review describes Maguid et al. 2006 and 2008 as using an anisotropic ENM; the papers themselves use experimental B-factors (2006) and the isotropic GNM (2008).

4 Eigenvector and eigenvalue changes

A mutation changes both the eigenvectors and the eigenvalues. To first order in $\delta\mathbf{K} = \mathbf{K}^{\text{mut}} - \mathbf{K}^{\text{wt}}$,

$$\delta\lambda_n = \mathbf{u}_n^T \delta\mathbf{K}\,\mathbf{u}_n, \qquad c_{mn} = \frac{\mathbf{u}_m^T \delta\mathbf{K}\,\mathbf{u}_n}{\lambda_n - \lambda_m} \quad (m \neq n): \tag{1}$$

eigenvalues respond to the diagonal of $\delta\mathbf{K}$ in the wild type's basis, eigenvectors to the off-diagonal, divided by the gaps.

Eigenvectors. `nhn` is the exponential of the entropy of the distribution O^2_{nm} of wild-type mode n over the mutant's modes: the number of mutant modes it spreads into. It needs no reassignment of modes.

Eigenvalues. Comparing the sorted spectra, μ_n against λ_n, is itself an assignment: it pairs modes by rank, which pairs different modes wherever they cross. Only functions symmetric in the eigenvalues, such as $\sum_n \ln \lambda_n$ in `ddg_tds`, are free of assignment. Two per-mode quantities are free of it:

$$\mathbf{u}_n^T \mathbf{K}^{\text{mut}} \mathbf{u}_n = \lambda_n + \delta K_{nn} = \textstyle\sum_m O^2_{nm} \mu_m, \tag{2}$$

$$\mathbf{u}_n^T \mathbf{C}^{\text{mut}} \mathbf{u}_n = \textstyle\sum_m O^2_{nm} / (\beta \mu_m). \tag{3}$$

Both are Rayleigh quotients at the wild-type mode: the eigenvector change enters them at second order. Their second-order contamination falls on opposite ends of the spectrum: the $\mathbf{C}$ form weights by $1/\mu$, so a small admixture of a soft mutant mode into a stiff wild-type mode can dominate it; the $\mathbf{K}$ form weights by μ, the reverse. How large either effect is on real mutants was not measured. `dmsfn` is the $\mathbf{C}$ form minus σ_n^2.

If the springs are perturbed in proportion to their stiffness, $\delta k_{ij} = \epsilon_{ij} k_{ij}$ with mean $\bar{\epsilon}$, then $E[\delta\lambda_n] = \bar{\epsilon}\lambda_n$ to first order: the relative change of the eigenvalues is the same for every mode, $\delta(1/\lambda) \propto 1/\lambda$, and the low modes change most in absolute fluctuation and equally in relative terms.

Close is not degenerate. When $\lambda_n = \lambda_m$ exactly, the individual eigenvectors are undefined within their subspace and their rotation has no physical meaning. When the eigenvalues are close but distinct, the eigenvectors are defined but sensitive: a perturbation with $|\delta K_{nm}| \gtrsim |\lambda_n - \lambda_m|$ rotates them, and the mutant's normal modes really are different. Closeness is the gap measured against the perturbation, not a property of the spectrum alone. When two close modes rotate within their span, the covariance changes little, by about the gap times the rotation: the vibrational directions can change a lot while the fluctuations change little. `nhn` measures the first; measures built on $\mathbf{C}$ measure the second.

5 A network and its complement

For a graph Laplacian $\mathbf{L}$ (the GNM's Kirchhoff matrix), the complement graph's Laplacian is $\bar{\mathbf{L}} = n\mathbf{I} - \mathbf{J} - \mathbf{L}$, with $\mathbf{J}$ the matrix of ones. Both annihilate $\mathbf{1}$, so they commute, share eigenvectors, and $\bar{\mathbf{L}}\mathbf{u} = (n - \lambda)\mathbf{u}$ for $\mathbf{u} \perp \mathbf{1}$: the complement's lowest modes are the graph's highest. With weights, the same holds for complement weights $c - w_{ij}$. On 2acy (`complement.R`, $d_{\max} = 10.5$ Å): $\mathbf{L} + \bar{\mathbf{L}} = n\mathbf{I} - \mathbf{J}$, the commutator is exactly zero, and the non-zero eigenvalues of $\bar{\mathbf{L}}$ are $n - \lambda$.

For the 3D Hessian of an ANM it fails: the complete network's Hessian has blocks $\mathbf{e}_{ij}\mathbf{e}_{ij}^T$ that depend on the geometry and is not a multiple of a projector. On 2acy the relative commutator $\|\mathbf{H}\bar{\mathbf{H}} - \bar{\mathbf{H}}\mathbf{H}\| / (\|\mathbf{H}\|\|\bar{\mathbf{H}}\|)$ is 0.017, and the best overlap of each mode of $\mathbf{H}$ with any mode of $\bar{\mathbf{H}}$ has median 0.22.

In the GNM, perturbing G by δ is exactly perturbing $\bar{G}$ by $-\delta$, so the robustness of mode n of G equals that of the mode at the opposite end of $\bar{G}$'s spectrum. This is a symmetry between two graphs, not between the low and high modes of one graph: a contact graph is sparse and its complement dense. What it does show is that low versus high is not intrinsic to eigenvector stability, which depends on $|\lambda_n - \lambda_m|$ and $|\delta K_{nm}|$, both unchanged by $\lambda \to c - \lambda$; the profile of robustness against n must come from how gaps and couplings are distributed over the spectrum.

6 A mutational scan: is mean nhn explained by the spectral density?

Setup. The genm vignette's wild type: `node = "ca"`, `model = "ming_wall"`, $d_{\max} = 10.5\,\text{Å}$, `d_max_graph = 14`, `ensemble = 1`, `alphabet_size = 10`, `mut_dl_sigma = 0.3`, `mut_sd_min = 2`. All 882 single-point mutants (98 sites, 9 letters each; `nhn_scan.R`). For each mutant: `nhn`, and $\delta\mathbf{K}$ in the wild type's internal mode basis, $\mathbf{U}^T(\mathbf{K}^{\text{mut}} - \mathbf{K}^{\text{wt}})\mathbf{U}$. Per-mutant `nhn` ranges over 1–22.

Models. `nhn` is estimated from two-level mixing: for each pair, $\tan 2\theta_{nm} = 2|\delta K_{nm}|/|\Lambda_n - \Lambda_m|$, weights $w_{nm} = \tan^2\theta_{nm}$ with $w_{nn} = 1$, normalised to a distribution whose exponential entropy is the estimate. The couplings come from different sources: (i) each mutant's own $\delta\mathbf{K}$, with $\Lambda = \lambda + \operatorname{diag}\delta\mathbf{K}$; (ii)–(v) random symmetric Gaussian couplings with $\Lambda = \lambda$, averaged over 40 draws (seed 1), with variance (ii) the scan's mean δK_{nm}^2 for each pair, (iii) one constant, (iv) one constant for each pair of spectral bands, (v) proportional to $\lambda_n\lambda_m$. Constants are fitted by least squares to the mean `nhn` profile (`nhn_explain.R`, `nhn_bands.R`).

The spectrum has two bands. λ jumps from 65 to 147 between modes 191 and 192; 97 modes lie above, the number of $i, i+1$ springs, which ming_wall makes 42 times stiffer. That the upper band is bond stretching is a guess from the count; the mode shapes were not checked.

Mean `nhn` has one hump per band (Figure 1): about 1 at the lowest modes, around the band gap ($n \approx 175$–225) and at the top, with maxima of 2.68 at $n = 89$ and about 2.2 near $n = 240$. With this model the profile is an M, not a single inverted U.

couplings δK_{nm}	fitted parameters	Pearson	Spearman
(i) each mutant's own	0	0.977	0.98
(ii) random, scan's mean square per pair	0	0.929	0.95
(iii) random, one constant (spectrum only)	1	0.803	0.82
(iv) random, one constant per pair of bands	3	0.867	0.87
(v) random, $\propto \sqrt{\lambda_n\lambda_m}$	1	0.486	0.69

Correlations are with the exact mean `nhn` over modes.

What the spectrum explains. With one constant coupling, the model reproduces the fine structure of the lower band: the peaks of `nhn` sit at small gaps. The fitted constant, 0.039, matches the measured rms coupling within the lower band, 0.038; model (iv) fits 0.197 for the upper band against 0.19 measured. Its cross-band scale, 1.02 against 0.218 measured, is poorly determined.

What it misses.

- The upper band couples about five times more strongly than the lower; with one constant coupling its hump disappears (Figure 2).
- The lowest modes, $n \lesssim 35$, are more robust than their gaps predict: their rms coupling to the other modes is about half the middle modes', rising over the first ∼20 modes as λ goes from 1 to ∼10. An unchecked guess at why: soft modes are near-rigid motions of domains that barely deform the local springs a mutation perturbs.
- The model undershoots around $n \approx 140$–175; not looked into.

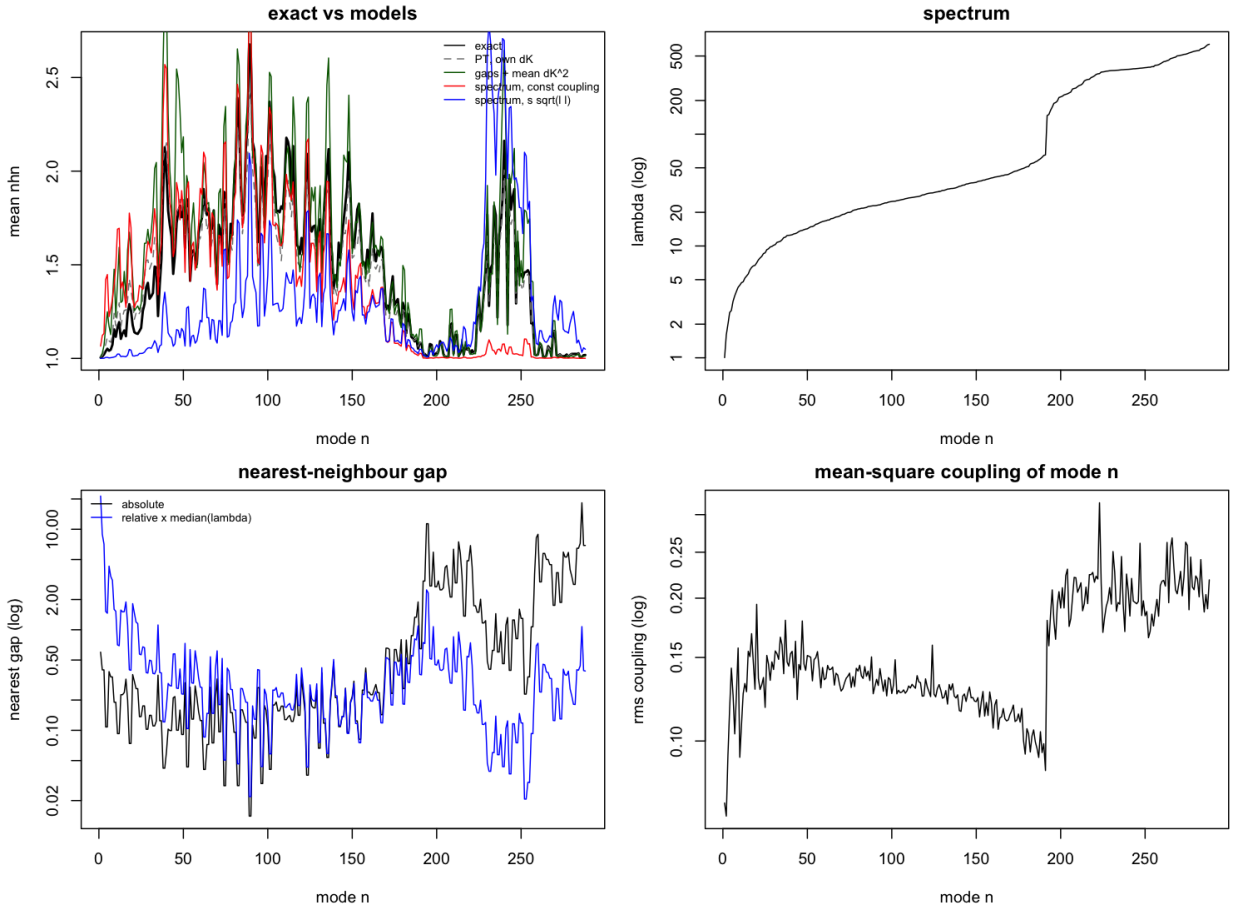


Figure 1: Top left: mean `nhn` over the 882 mutants (black) and the models (i), (ii), (iii) and (v). Top right: the wild type's spectrum. Bottom left: the gap to the nearest eigenvalue, absolute and relative. Bottom right: the root-mean-square coupling of mode n to the other modes, over the scan.

So the spectral density accounts for most of the profile, including where its peaks are, but not all of it: the lowest modes are robust both because of their gaps and because they couple weakly to mutations, and the second is not a spectral effect.

7 Open questions

- With a single band, `model = "anm"`, is the profile a single inverted U?
- Does the lowest modes' coupling to mutations scale with λ?
- How large is the second-order contamination of the two Rayleigh quotients on real mutants?
- Do these results hold for other proteins, models and ensembles?

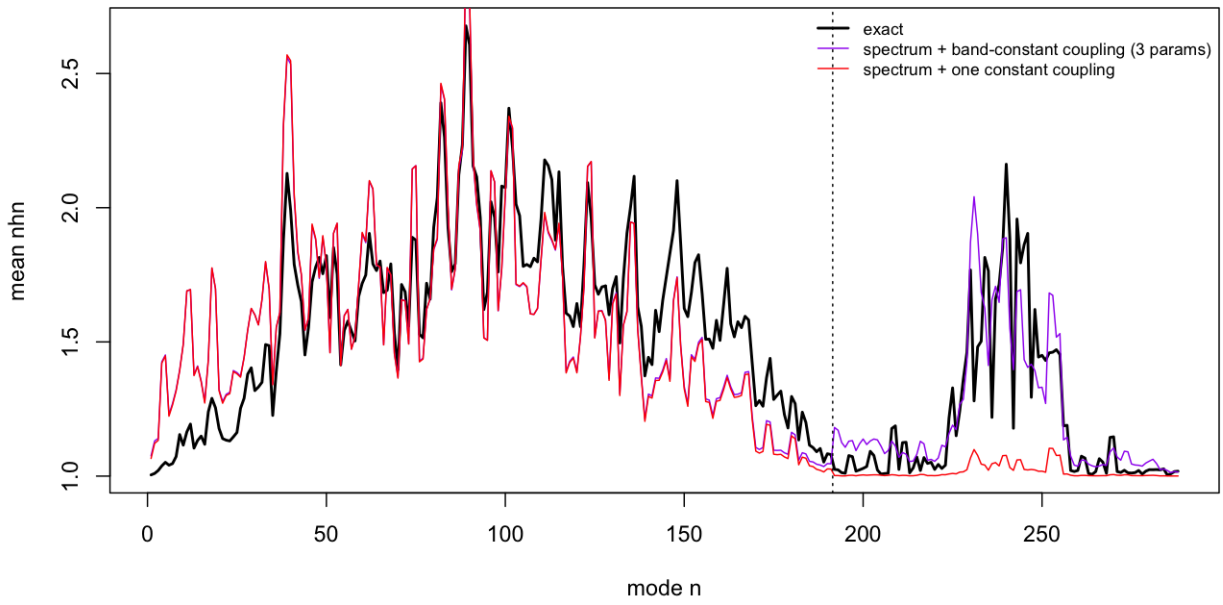


Figure 2: Mean `nhn` (black), with model (iv), one coupling scale per pair of bands (purple), and model (iii), one constant coupling (red). The dotted line marks the band gap.